BUSINESS REQUIREMENTS DOCUMENT

ERP Sample Project

Feature Summaries
Prepared by Kolosal AI

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BUSINESS REQUIREMENTS DOCUMENT

Product List Page

Prepared by Kolosal AI

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1. Frontend

1.1 Overview

The Product List Page provides a centralized view for users to manage and browse the company's product catalog. It allows users to quickly locate specific items using search and filtering tools, view key product details in a structured table, and easily navigate to create new product entries.

1.2 Step-by-Step Flow

1. Initial Load: When the page is first opened, the system automatically retrieves the full list of products and displays them in the table.
2. Filtering: Users can enter search terms or select a specific brand from the filter bar to narrow down the results.
3. Applying Search: Once the desired filters are set, the user clicks the "Apply Search" button to refresh the table with the filtered results.
4. Viewing Details: Users can click on any product name in the table to be redirected to a dedicated page containing more comprehensive information about that specific item.
5. Adding Products: A dedicated button is available to redirect the user to the "Add Product" page to create new entries in the system.

1.3 Key Concepts

- Filter Bar: A set of controls that allows users to refine the product list by specific criteria, such as product name or brand, to find items more efficiently.
- Product Table: A structured display showing essential product information, including the SKU, name, brand, internal product code, tax rate, and price.
- Navigation: The ability to move between the main list view, individual product detail pages, and the product creation form.
- Data Refresh: The process of updating the table content based on the user's search inputs without needing to reload the entire page.

1.4 Business Value

This feature improves operational efficiency by providing a streamlined interface for inventory management. By enabling quick searches and clear visibility into product data (such as pricing and tax information), staff can spend less time looking for information and more time managing the catalog. The intuitive navigation ensures that users can move seamlessly between browsing, reviewing details, and adding new inventory.

2. Logic

2.1 Overview

The Product List Page feature allows users to efficiently browse, search, and filter the company’s product catalog. It provides a structured view of all available items, ensuring that users can quickly locate specific products by name, code, or brand. The system is designed to handle large inventories by displaying results in manageable segments, ensuring the page remains fast and easy to navigate.

2.2 Step-by-Step Flow

1. Request Initiation: When the page loads or a user clicks the "Apply Search" button, the system captures the current search terms, brand filters, and page number.
2. Data Retrieval: The system searches the product database for items that match the user's criteria (specifically looking for matches in the product name or product code) and applies the selected brand filter.
3. Sorting: The retrieved products are automatically sorted so that the most recently added items appear at the top of the list.
4. Pagination: The system calculates which specific "slice" of the data to show based on the requested page number and the limit of items per page.
5. Display: The system maps the product details—such as SKU, Name, Brand, Code, Tax, and Price—to the corresponding fields on the screen for the user to view.

2.3 Key Concepts

- Pagination: A method of breaking a large list of products into smaller, numbered pages to improve performance and readability.
- Filtering: The ability to narrow down the product list by specific criteria, such as brand, to help users find exactly what they need.
- Search Query: A text-based input that allows users to find products by typing in a name or a specific product code.
- Data Mapping: The process of taking raw information from the database and organizing it into the specific labels and fields shown on the user interface.

2.4 Business Value

This feature streamlines the user experience by reducing the time spent searching for products. By providing robust filtering and organized pagination, it enables staff to manage inventory more effectively and respond to customer inquiries faster. The consistent display of pricing and tax information ensures that users have accurate, up-to-date data at their fingertips, supporting better decision-making and operational efficiency.

3. Data

3.1 Overview

The Product List Page feature provides a centralized way for users to view, search, and navigate the company's product catalog. It is designed to handle large volumes of data efficiently by allowing users to narrow down results using specific search terms or brand filters, while presenting the information in manageable, paginated chunks.

3.2 Step-by-Step Flow

1. Request Processing: The system receives your search criteria (such as a keyword or brand name) and your current page location.
2. Data Filtering: The system searches the product database to find items that match your specific keywords (by name or SKU) and filters them by the selected brand.
3. Counting Results: The system calculates the total number of products that match your criteria to ensure the page navigation accurately reflects how many pages of results are available.
4. Pagination: The system retrieves only the specific subset of products required for your current view, ensuring the page loads quickly even if there are thousands of items.
5. Data Formatting: The system organizes the raw product information into a clean, readable format, including key details like the SKU, name, brand, pricing, and tax information.
6. Display: The final list of products is presented on your screen, ready for review.

3.3 Key Concepts

- Filtering: The ability to narrow down a large list of products by entering a search term or selecting a specific brand.
- Pagination: A method of breaking a large list of items into smaller, separate pages to improve performance and readability.
- SKU (Stock Keeping Unit): A unique identifier used to track individual products within the inventory.
- Metadata: Information about the search results (such as the total count) that helps the system manage how many pages of data are available to the user.

3.4 Business Value

This feature improves operational efficiency by allowing staff to quickly locate specific items within the ERP system. By combining search and filtering capabilities with pagination, the system remains fast and responsive, reducing the time spent searching for product information and ensuring that users can easily access the exact data they need to manage inventory or assist customers.

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Products Menu

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1. Frontend

1.1 Overview

The Products Menu is a navigation component designed to help users easily access and manage product-related information within the ERP system. It features a clean, organized layout that uses an "accordion" style—a collapsible list—to keep the interface uncluttered while providing quick access to specific areas like Inventory, Categories, and Suppliers.

1.2 Step-by-Step Flow

1. Initial View: When you first open the ERP system, the "Products" menu is visible and expanded by default, showing its sub-categories (Inventory, Categories, and Suppliers).
2. Collapsing/Expanding: You can click on the "Products" header at any time to hide or show the sub-menu list. A small arrow (chevron) icon indicates whether the list is currently open or closed.
3. Selecting Items: Clicking on any specific sub-menu item (such as "Inventory") or other main navigation items (such as "Reports") will highlight that selection, letting you know exactly where you are in the system.
4. Navigation: Once an item is selected, the system updates the main view to display the relevant page for that specific section.

1.3 Key Concepts

- Accordion Menu: A design pattern that allows a section of content to be expanded or collapsed, helping to save screen space.
- Active State: A visual indicator (highlighting) that shows the user which menu item is currently selected or "active."
- Chevron: A small arrow icon used to signal that a menu item can be expanded or collapsed.
- Navigation Item: A clickable link that directs the user to a specific area or page within the ERP software.

1.4 Business Value

This feature improves the user experience by providing a predictable and intuitive way to navigate complex product data. By grouping related tasks under a single "Products" menu, users can find what they need faster, reduce visual clutter on their screens, and maintain a clear sense of their current location within the application. This streamlined approach helps increase productivity by minimizing the time spent searching for specific management tools.

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Main Menu

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1. Frontend

1.1 Overview

The Main Menu is the primary navigation header for the ERP system. It acts as the central hub for users to move between different areas of the application, manage their personal account settings, view system alerts, and perform quick searches across the platform.

1.2 Step-by-Step Flow

1. Navigation: Users click on top-level menu items (such as "Dashboard" or "Products") to jump to specific sections of the ERP. The system highlights the active tab to show the user their current location.
2. Product Exploration: Clicking the "Products" menu item reveals a dropdown list, allowing users to access specific product-related features or categories.
3. Searching: Users can type keywords into the search bar. The system waits for the user to finish typing before processing the request, which ensures a smooth and efficient search experience.
4. Notifications: Clicking the notification bell icon toggles a dropdown menu that displays recent system alerts or updates.
5. Account Management: Clicking the user avatar opens a profile menu, providing quick access to account settings and the option to sign out of the system.

1.3 Key Concepts

- Navigation Items: The primary links used to move between different functional areas of the ERP.
- Dropdown Menus: Contextual lists that appear only when requested (e.g., for Products, Notifications, or Profile settings) to keep the interface clean.
- Debounced Search: A performance feature that delays the search action until the user has stopped typing, preventing the system from performing unnecessary work while the user is still entering text.
- Active State: A visual indicator that identifies which section of the ERP the user is currently viewing.

1.4 Business Value

This header component provides a consistent and intuitive user experience across the entire ERP system. By centralizing navigation, search, and account management, it reduces the time users spend looking for tools and helps them stay focused on their tasks. The use of "debounced" search and organized dropdowns ensures the interface remains responsive and uncluttered, even as the system grows in complexity.

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Products Sub Menu

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1. Frontend

1.1 Overview

The Products Sub Menu is a navigation component designed to help users easily switch between different product-related views within the ERP system. It functions as an accordion-style menu, providing a clean and organized way to access the main "Products" list and the "Service Products" list. By default, the system highlights the "Products" section when the user first arrives, ensuring they are immediately directed to the primary product view.

1.2 Step-by-Step Flow

1. Initialization: When the component loads, the system automatically sets "Products" as the active tab, ensuring the user sees the standard Product List Page by default.
2. User Interaction: The user selects either "Products" or "Service Products" from the navigation menu.
3. State Update: The system updates the visual interface to highlight the selected menu item, providing clear feedback on which section is currently active.
4. Navigation: The system redirects the user to the corresponding page (either the Product List or the Service Products List) based on their selection.

1.3 Key Concepts

- Accordion Menu: A compact navigation style that organizes links into a vertical list, allowing users to switch between related categories efficiently.
- Active Tab State: A background mechanism that tracks which menu item the user is currently viewing, allowing the system to highlight that item visually.
- Default Routing: The automatic behavior that ensures the user is placed on the most relevant page (the main Products list) as soon as the component is accessed.

1.4 Business Value

This feature improves the user experience by providing a predictable and intuitive way to navigate the ERP's product catalog. By centralizing these navigation options, users can switch between standard products and service-based products without leaving the context of the product management area. This reduces navigation time and helps users maintain focus on their tasks within the ERP system.

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GET /api/products

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1. Logic

1.1 Overview

This feature provides a way for authorized users to retrieve a list of all products stored in the ERP system. To ensure the system remains fast and easy to navigate, the product list is delivered in smaller, manageable chunks (pagination). Each request includes the product details—such as name, SKU, price, and creation date—along with helpful summary information like the total number of products available.

1.2 Step-by-Step Flow

1. Authentication Check: The system first verifies that the user is authorized to access the product data. If the user is not logged in or lacks permission, the request is denied.
2. Parameter Validation: The system checks the requested page number and the number of items per page. If these numbers are invalid (e.g., negative numbers), the system rejects the request.
3. Data Retrieval: The system calculates which specific products belong on the requested page and fetches them from the database, sorted by the date they were added.
4. Metadata Calculation: The system counts the total number of products in the database to help the user understand the full scope of the list.
5. Response Delivery: The system returns the requested list of products along with the pagination details. If no products are found for the requested page, or if a system error occurs, the user is notified accordingly.

1.3 Key Concepts

- Pagination: A method of breaking a large list of items into smaller, numbered pages to improve performance and readability.
- Query Parameters: Specific settings provided by the user (like "page number" or "items per page") that tell the system exactly which slice of data to return.
- Metadata: Additional information provided alongside the product list, such as the total count of items, which helps the user navigate the data effectively.
- Authentication: A security step that ensures only verified users can view sensitive business information.

1.4 Business Value

This feature enables efficient data management by allowing users to browse the product catalog without overwhelming the system or the user interface. By providing structured, paginated data, it ensures that staff can quickly locate specific items, track inventory growth via the total count, and maintain a smooth, responsive experience when interacting with the product database.

2. Data

2.1 Overview

The Product API feature allows users to retrieve a comprehensive list of all products currently stored in the ERP system. To ensure the system remains fast and easy to navigate, the feature includes "pagination," which breaks the product list into smaller, manageable chunks rather than loading everything at once.

2.2 Step-by-Step Flow

1. Request: The system receives a request for the product list, including optional settings for which page of results to view and how many items should appear on that page.
2. Data Retrieval: The system calculates which specific products belong on the requested page and fetches them from the database, sorted by the most recently created items first.
3. Counting: The system simultaneously counts the total number of products available in the database to help the user understand the full scope of the inventory.
4. Response: The system returns the requested page of products along with "metadata"—information that tells the user which page they are on, the limit per page, and the total number of products available.

2.3 Key Concepts

- Pagination: A method of dividing a large list of data into smaller pages to improve performance and readability.
- Metadata: Supplemental information provided with the product list (such as total count and current page number) that helps the user or application understand the context of the data returned.
- Sorting: The process of organizing the product list by date, ensuring that the newest items appear at the top of the results.

2.4 Business Value

This feature provides a reliable and efficient way for users to browse the product catalog. By implementing pagination, we ensure that the application remains responsive even as the product database grows significantly over time. Providing total count metadata also allows for better user interface design, such as enabling "jump to page" functionality or displaying progress indicators for the user.

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GET /api/products?utf8= &q[attribute_predicate]=value&...&page_rows=

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1. Logic

1.1 Overview

The Product Search API allows users to retrieve a filtered list of products from the catalog. It provides a flexible way to search by specific identifiers (like SKU or product code), categories (like brand or tax status), or custom descriptive fields. The feature is designed to be secure, ensuring that only authorized users can access the data, and it includes built-in validation to ensure that search criteria are accurate and exist within the system.

1.2 Step-by-Step Flow

1. Security Check: The system first verifies that the user is logged in and has the necessary permissions to view the product list.
2. Parameter Processing: The system collects the search criteria (such as name, brand, or tags) and the requested number of results per page from the user's request.
3. Validation: The system checks that any specific IDs provided in the search (such as a specific brand or tag ID) actually exist in the database. If an ID is invalid, the request is stopped.
4. Search Execution: The system applies the valid filters to the product database. This includes matching text fields, filtering by specific categories, and linking to product tags.
5. Result Delivery: The system returns the matching list of products to the user. If any errors occur during this process (such as invalid data formats), the system provides a clear error message.

1.3 Key Concepts

- Filtering: The ability to narrow down a large list of products by specific attributes like SKU, name, brand, or custom information fields.
- Pagination: A control mechanism that allows the user to specify how many products should appear on a single page of results.
- Validation: A safety step that ensures the search criteria are formatted correctly and that any referenced items (like a specific brand) are real entries in the system.
- Authorization: A security gate that confirms the user has the right to access product information before the search is performed.

1.4 Business Value

This feature empowers users to quickly and accurately locate specific items within a large inventory. By supporting a wide range of search criteria—from standard identifiers to custom information fields—it reduces the time spent manually browsing the catalog. Furthermore, the built-in security and validation ensure that the system remains reliable, secure, and provides consistent, high-quality data to the end user.

2. Data

2.1 Overview

This feature provides a flexible search interface for the product catalog. It allows users to retrieve a list of products by applying various filters, such as searching by name, SKU, or product code, as well as filtering by specific categories like brand, type, tax status, or assigned tags. The system also supports sorting the results and controlling the number of items returned per page to ensure efficient data retrieval.

2.2 Step-by-Step Flow

1. Receive Request: The system receives a search request containing various filter criteria, sorting preferences, and a limit for the number of results.
2. Apply Text Filters: The system searches for partial matches on fields like SKU, name, product code, and custom information fields.
3. Apply Exact Match Filters: The system narrows down results based on specific identifiers for product brands, product types, and tax categories.
4. Apply Relational Filters: If tags are provided, the system filters the product list to include only those associated with the selected tags.
5. Apply Sorting: The system organizes the resulting list based on the requested sort order (e.g., by name in ascending order).
6. Apply Pagination: The system limits the total number of results returned to the user based on the requested page size.
7. Return Results: The filtered, sorted, and paginated list of products is returned to the user.

2.3 Key Concepts

- Text Containment: A search method that finds items where a specific field contains the provided text (e.g., searching "Blue" will return "Blue Shirt" and "Light Blue Pants").
- Exact Match: A filter that requires the product attribute to be identical to the provided value (e.g., filtering by a specific Brand ID).
- Relational Filtering: A method of filtering products based on their connection to other data entities, such as tags.
- Pagination: A technique used to break a large list of results into smaller, manageable chunks to improve performance and readability.
- Sorting: The ability to arrange the final list of products in a specific order, such as alphabetical or numerical.

2.4 Business Value

This feature empowers users to quickly locate specific items within a large product catalog without needing to browse through every entry. By providing granular control over search criteria—such as filtering by brand or tag—the system improves operational efficiency for staff managing inventory or customer service representatives assisting clients. Additionally, the inclusion of pagination ensures that the application remains fast and responsive, even as the product catalog grows.

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Product Schema

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1. Data

1.1 Overview

The Product Schema feature provides a unified way to access and manage comprehensive product information within the ERP system. Instead of pulling data from multiple disconnected sources, this feature consolidates product details—including store-specific stock, warehouse inventory, branding, and categorization—into a single, structured record. This ensures that whenever you view, create, or update a product, you are working with a complete and consistent set of data.

1.2 Step-by-Step Flow

1. Request Initiation: When a user requests product details (or performs a search), the system triggers a consolidated data query.
2. Data Aggregation: The system automatically joins information from several areas:

* Core Product Data: Basic details like name, SKU, and description.

* Categorization: Links to specific product brands and types.

* Inventory Levels: Real-time stock counts from both store locations and warehouses.

* Metadata: Associated tags and historical audit logs.

3. Calculation: The system calculates "available quantity" by subtracting reserved and consignment stock from the total inventory on hand.
4. Response Formatting: The gathered data is structured into a standardized JSON format, ensuring the information is ready for the user interface or other system processes.
5. Pagination: For lists of products, the system applies limits and offsets to ensure the application remains fast and responsive, even with large catalogs.

1.3 Key Concepts

- Consolidated Query: A single database operation that merges information from multiple tables (like brands, types, and inventory) to provide a "360-degree view" of a product.
- Available Quantity: A calculated value representing stock that is truly ready for sale, derived by taking total inventory and removing items that are reserved or held on consignment.
- SKU (Stock Keeping Unit): The unique identifier used to track specific products across all stores and warehouses.
- Audit Trail: A historical record of changes made to a product, such as price updates or status changes, which allows for tracking and accountability.
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Pagination: A method of breaking large lists of products into smaller, manageable pages to improve system performance and user experience.

1.4 Business Value

- Single Source of Truth: By consolidating data, you eliminate discrepancies between store stock and warehouse inventory, ensuring everyone sees the same accurate numbers.
- Improved Decision Making: With instant access to pricing, stock levels, and product history, managers can make faster, data-driven decisions regarding inventory replenishment and sales strategies.
- Operational Efficiency: The system handles complex calculations (like available stock) automatically, reducing manual work and the risk of human error.
- Scalability: The use of pagination and structured data ensures that the system remains fast and reliable as your product catalog and store count grow.

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Common Standard Response JSON Schema

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1. Data

1.1 Overview

This feature establishes a standardized "language" for how our ERP system communicates back to users or other software. By creating a consistent structure for every response—whether it is a successful data retrieval or an error message—we ensure that the system behaves predictably. This uniformity makes it easier for developers to build reliable integrations and ensures that users receive clear, consistent feedback when something goes right or wrong.

1.2 Step-by-Step Flow

1. Request Processing: When a user or system interacts with the ERP, the system processes the request.
2. Outcome Determination: The system identifies the result of the request (e.g., success, validation error, permission issue, or server error).
3. Standardized Formatting: Based on the outcome, the system applies a pre-defined "template" (schema) to the response.
4. Consistent Delivery: The system sends the response back in a predictable format, ensuring that key information—such as the status code, error details, or success data—is always located in the same place.

1.3 Key Concepts

- Standardized Response: A uniform structure for all API communications, ensuring that every message follows the same rules for layout and content.
- Status Codes: Numerical indicators that tell the user exactly what happened (e.g., "201" for success, "404" for not found, or "500" for a server error).
- Schema: A blueprint or set of rules that defines exactly what information must be included in a response, such as error messages or specific data fields.
- Validation Errors: Specific feedback provided when a request is formatted correctly but contains information that the system cannot process (e.g., missing required fields).

1.4 Business Value

- Improved Reliability: By enforcing a standard format, we reduce the likelihood of "surprises" or crashes when different parts of the software communicate with each other.
- Faster Development: Developers spend less time guessing how to handle responses because every endpoint follows the same rules, allowing them to build new features more quickly.
- Better Troubleshooting: Because error messages are standardized, support teams and developers can identify and resolve issues much faster, as the system provides consistent, predictable diagnostic information.
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Easier Integrations: External partners or third-party tools connecting to our ERP will have a much easier time reading our data, as they only need to learn our standard format once.

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Product

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1. Data

1.1 Overview

The Product feature serves as the central hub for managing your inventory catalog within the ERP system. It provides a structured way to define products, categorize them by brand and type, and track their movement, pricing, and history across different physical locations like stores and warehouses. This system ensures that every item is uniquely identified and that all changes—whether in stock levels or pricing—are recorded for audit and reporting purposes.

1.2 Step-by-Step Flow

1. Product Definition: Every item is registered in the central product catalog with a unique identifier (SKU), name, and classification (Brand and Type).
2. Categorization & Tagging: Products are organized using predefined brands and types, and can be further labeled with custom tags for easier searching and reporting.
3. Inventory Management: The system tracks stock levels separately for stores and warehouses. Any movement of stock (such as sales or transfers) is automatically logged in a history table to maintain an accurate audit trail.
4. Pricing Control: The system supports standard pricing as well as flexible "override" pricing, allowing you to set specific rates based on the region, store, warehouse, or customer group.
5. Historical Tracking: Every significant change to a product—including price updates, status changes, and procurement history—is captured in dedicated history logs, ensuring full transparency over the product's lifecycle.

1.3 Key Concepts

- SKU (Stock Keeping Unit): A unique code assigned to each product to ensure it is easily identifiable and distinct from all other items.
- Lookup Tables: Standardized lists (like Brand and Product Type) used to ensure consistent categorization across the entire system.
- Stock Audit Trail: A chronological record of all inventory changes, which helps in identifying when and why stock levels increased or decreased.
- Price Overrides: The ability to apply different pricing rules to a product depending on the context, such as a special price for a specific store or a wholesale customer group.
- Product Lifecycle History: A comprehensive log that tracks changes to a product's status (e.g., active/inactive) and pricing over time, providing a clear view of how a product has evolved.

1.4 Business Value

- Operational Accuracy: By centralizing product data and automating stock tracking, you reduce the risk of manual errors and inventory discrepancies.
- Informed Decision Making: With detailed purchase and price history, you can easily analyze procurement costs and pricing trends to optimize your margins.
- Scalability: The flexible structure allows you to manage complex pricing strategies across multiple locations and customer segments without needing to create duplicate product entries.
- Full Accountability: The built-in audit logs for stock movements and price changes provide complete visibility, making it easier to manage compliance and internal reporting.